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Consumer Product Safety Analysis

1. Business Understanding

1.1 Background

1.1.1 Organizational/Context Description

The Consumer Safety Research Institute (CSRI) is a specialized fictional nonprofit organization established in 2012 to perform independent research on consumer product safety and publish reports and insights that defines public disclosure, and aids in policy decisions. Consumer Safety Research Institute collects comprehensive injury data through systems like National electronic Injury surveillance system (NEISS) – which is a national database managed by U.S Consumer Product Safety Commission (CPSC) to track injuries treated in hospital emergency departments across the United States. While government agencies collect essential data, their raw reports are often cumbersome for the average consumer to understand and comprehend. CSRI fills this critical void, transforming complex datasets into clear public understanding of product risks.

The CSRI publishes an annual "Product Safety Spotlight" report that examines the emerging trends in product safety, identify areas of public attention and provides recommendations for consumers, manufacturers and policymakers. The report serves multiple stakeholders like the parent advocacy groups, policy researchers, product liability attorneys, insurance industry and regulatory agencies.

1.1.2 Domain and Context Assessment

The domain of consumer product safety is currently characterized by a significant disconnect between available data and public

understanding. While the National Electronic Injury Surveillance System (NEISS) provides a robust foundation of comprehensive injury statistics, this raw data is technically complex and largely inaccessible to non-specialists. Consequently, the prevailing public discourse frequently relies on anecdotal evidence, high-profile media incidents, or biased manufacturer marketing rather than systematic, empirical analysis. This reliance on incomplete information leaves consumers, policymakers, and safety advocates without the evidence-based insights necessary for informed decision-making.

This information gap creates a critical challenge in an environment where stakeholders increasingly demand rigorous, independent verification to distinguish between isolated accidents and genuine systemic risks. There is a pressing market need for an authoritative intermediary capable of translating dense surveillance data into clear, actionable intelligence. However, few existing organizations possess the requisite combination of technical data expertise and institutional independence to conduct this analysis effectively.

The Consumer Safety Research Institute (CSRI) is strategically positioned to bridge this divide. By leveraging national injury surveillance data, CSRI transforms abstract statistics into evidence-based answers regarding product safety trends. The institute's analytical work serves to ground public debate in fact, shifting the landscape from reactionary responses toward data-driven policy. In doing so, CSRI not only empowers consumers and regulators with clarity but also proactively identifies emerging safety threats, ensuring that manufacturers and governing bodies can address priority issues with precision and authority.

1.1.3 Problem Situation

While the government collects vast amounts of injury data, there is no effective mechanism to translate this raw information into accessible knowledge for the public.

Without independent analysis, safety decisions are made based on fear, marketing, or anecdotes rather than statistical reality, potentially leaving dangerous products on the market or causing unnecessary panic over minor issues.

There is an urgent requirement for an independent body capable of analyzing complex surveillance data and communicating objective, actionable findings to non-technical stakeholders.

1.1.4 Data Science Justification

Consumer Safety Research Institute (CSRI)has visibility of National Electronic Injury Survey Data (NEISS) data from 2005 through 2024. The availability of NEISS transactional (at injury level) and demographic (Age/Sex//Race) data of 19 years represents a massive, high dimensional repository of public injury information. The volume and complexity of this data make any manual analysis cumbersome and time consuming.

We need to apply advanced machine learning and statistical analysis on this data to gather below insights from this massive dataset.

1.1.4.1 Trend Analysis and Seasonality

1.1.4.2 Identify Unstructured Narrative Data

1.1.4.3 Demographic Clustering

1.1.4.4 Anomaly Detection

1.1.4.5 Identify correlations between predictors causing Injury

1.1.4.6 Complex survey Data and the impact on Statistical Inference.

1.1.4.7 Identify predictors causing hospitalization due to injuries.

1.1.5 Stakeholder Identification

The CSRI organization helps safety experts understand the emerging hazards caused due to consumer product usage. They use this knowledge to verify the danger and protect public from injuries related to consumer products.

The report generated by CSRI also aids the safety and product engineers perform an in-depth analysis of the root cause for the injuries and then perform a risk assessment where they compare the severity of the injury to the probability of the risk and then identifies the vulnerable population.

The reports and insights generated by CSRI is also used by experts and government agencies to issue product recalls and to rewrite the safety rules that the manufactures need to follow. The knowledge gained from these reports are also broadcasted for public disclosure to prevent future injuries in the form of safety alerts.

The findings in these reports are also published to warn manufacturers, retailers, insurers and policy makers that a new regulation is needed to prevent injuries.

1.2 Project Objectives and Success Criteria

1.2.1 Primary Objective

Develop predictive models to identify which consumer product injuries are most likely to result in hospitalization and determine what factors—product categories, demographics, injury characteristics—predict severe medical outcomes among emergency department cases, producing evidence-based findings that inform CSRI's 2025 Product Safety Spotlight report and

contribute to public understanding of consumer product safety priorities.

1.2.2 Secondary Objectives

Identify temporal and demographic trends in consumer product injury severity patterns, including seasonal variations, changes over time, and emerging patterns that may indicate shifting safety concerns, generate insights about demographic patterns in severe product injuries that inform targeted safety messaging, and create analytical approaches that enhance CSRI's reputation as a leading source of independent consumer safety research.

1.2.3 Quantitative Success Criteria

1.2.3.1 Model Performance

Predictive Accuracy: The predictive models should achieve a Receiver Operating Characteristic (ROC) AUC score of > 0.70 thereby indicating an ability to strongly distinguish between severe (hospitalized) vs non-severe cases (not hospitalized).

Statistical Significance: The predictors identified from the statistical analysis must demonstrate statistical significance with $p < 0.05$ to be considered valid predictors.

Goodness of Fit: The model must pass standard calibration tests to ensure that predicted probabilities reasonability match observed hospitalization.

1.2.3.2 Trend Analysis

The analysis must successfully quantify seasonal variations (identify peak injury months) and long-term trends (annualized percentage change). The analysis should also identify at least 3-5 emerging injury patterns that signals shifting safety concerns.

1.2.3.3 Output Deliverables

The project must generate at least 3-5 visualizations and 3 summary tables for the inclusion in the CSRI 2025 Safety Spotlight Report

1.2.4 Qualitative Success Criteria

1.2.4.1 Actionability of Insights

The insights gathered from the statistical analysis should be specific enough to be translated into actionable safety advice.

1.2.4.2 Model Interpretability and Explainability

The factors causing hospitalization should be explainable to non-technical audiences and should not be a black box.

1.2.4.4 Organizational Impact

The analytical approach should be robust enough to withstand the scrutiny of external reviewers and industry critics thereby enhancing CSRI's reputation.

The analytical approaches should be documented and reproducible.

The work must meet the institute's standards for scientific rigor and public communication.

The analysis will be published under CSRI's name and must withstand scrutiny from diverse stakeholders including consumer advocates, industry representatives, academic researchers, and policy makers.

1.2.5 Impact Assessment

The report will provide Consumer advocacy groups and policymakers data-backed evidence to support safety recommendations and regulatory decisions.

By systematically analyzing injury patterns, the report will contribute to more informed public discourse while identifying emerging safety issues that warrant attention from consumers, manufacturers, and regulators.

1.2.6 Timeline and Milestones

Expect to complete the statistical analysis within 8 weeks, draft findings within 10 weeks and publish the final article within 12 weeks of project start.

1.3 Assessment of Responsibility

1.3.1 Legal and Regulatory Considerations

The data that is available through NEISS is a public domain information and will not need any copyrights to use the information.

This data does not have any Personally Identifiable information like the name, address, SSN, Manufacturer or Hospital information.

We should not make any attempts during /after the project life cycle to re-identify the persons, hospitals, manufactures by linking this data to other external datasets/vectors.

CPSC is bound by the Consumer Product Safety Act (CPSA) Section 6(b) and Privacy Act of 1974 that strictly regulates the release of information that could identify individuals or manufacturers without a fair chance to respond.

CPSC requires that any public analysis of the data acknowledges the source and include a disclaimer "This analysis is based on data from the U.S. Consumer Product Safety Commission (CPSC) but is not an official determination or policy of the CPSC." (Needs to be added in the Phase-6 Deployment Section)

1.3.2 Privacy and Data Protection

Patient Anonymity:

The NEISS data is available for public consumption and is deidentified to remove any PII comply with the Privacy Act of 1974 and HIPPA Act.

The data is devoid of any direct identifiers (Name, Address, Phone Numbers, SSN) before released. The Treatment Date in the dataset is preserved to maintain the seasonality but the time in the data omitted or generalized/shifted to prevent direct/indirect identification from the police report or ambulance log. The narratives are also scrubbed to remove any PII information from creeping in. We should not make any attempts during /after the project life cycle to re-identify the persons.

Hospital Confidentiality:
The identities of 100 participating hospitals are kept confidential, but NEISS releases a geographical map which helps us to identify were/which state and the strata of the hospitals participating the program. No hospital analysis is possible using the NEISS dataset for this project.

Manufacturer Protection:
The consumer product safety act (Section 6(b)) restricts CPSC from identifying any manufacturers in the data. Brand/Manufacturer information in not included in the NEISS dataset and hence we need to summarize the analysis based on product categories.

Statistical Disclosure Limitation:
Rare events or data could help in re-identification like (110-year-old male injured by a product on a specific day). CPSC rule needs us to follow cell suppression rules if a query yields fewer than 20 cases nationwide.

1.3.3 Bias and Fairness Concerns

1.3.3.1 Selection Bias

The NEISS dataset includes data for injuries that were reported to the Emergency Departments and does not include the information of people going to the primary care physicians, urgent care or individuals taking self-care. This data also does not include information of people who died at the scene due to a product usage as they are taken directly to the morgue.

This introduces a fairness concern as lower income people might not have a PCP and could always use the ED for non-emergencies. This could inflate the injury rates for people with lower income while undercounting injuries of people with higher income.

1.3.3.2 Reporting Bias

The data is manually entered by hospital NEISS hospital coordinators using the data entered by doctors and nurses into the patients' medical record. This data could have been entered hurriedly, and this could possibly introduce reporting bias whereby the data could have many unknown values. If models drop rows with NULL/UNKNOWN data, we could exclude data that could have been useful.

1.3.3.3 Socio Economic Bias

Exposure to injury also depends on socio economic background as the exposure to products is not random. If the analysis only focuses on the number of injuries without adjusting for the exposure (number of people who owning the product) then It could cause a bias.

1.3.3.4 Language Bias

The narratives against the NEISS case are written in English. If the patient doesn't know English, they will not be able to effectively communicate how injury happened to the primary care givers or to the nurse/doctors.

1.3.3.5 Geographic Bias

The NEISS data is not collected for every hospital in the USA. NEISS is a probability sample, and it generalized the hospitals into 100 buckets. This could potentially mask localized issues.

1.3.5 Stakeholder Impact Analysis

Positive impacts include informed residents making better product choice and evidence-based policy making; negative impacts could include property value changes in affected neighborhoods, stigmatization of certain communities, or misinterpretation leading to unnecessary public alarm

1.3.5 Mitigation Strategies

Use statistical methods that account for monitoring bias; present findings with appropriate uncertainty ranges; include community context in reporting; establish fact-checking protocols for public-facing analysis.

1.4 Data Science Goals & Success Criteria

1.4.1 Technical Problem Framing

1.4.2 Data science Objectives

1.4.3 Technical Success Criteria

1.4.3.1 Development of statistically sound predictive models that can reliably identify factors associated with hospitalization for product-related injuries.

1.4.3.2 Clear identification of meaningful patterns in injury severity across different product categories and demographic groups.

1.4.3.3 Rigorous analysis that produces findings robust enough to support evidence-based safety recommendations.

1.4.3.4 Effectively situating insights from the analysis to contribute new understanding to consumer product safety discourse and inform practical safety guidance.

1.4.3.5 Effective communication of technical findings to diverse audiences including consumers, policymakers, and safety advocates through accessible analysis and clear visualizations.

1.4.4 Analytical Approach Overview

1.4.5 Objective-Technical Mapping

1.4.6 Constraints and Assumptions

1.5 Project Plan

1.5.1 Phase specific planning

1.5.2 Timeline and milestones

1.5.3 Risk identification and contingencies

2 Data Understanding

2.1 Data Inventory

2.1.1 Primary Dataset Description

The primary dataset used by the Consumer Safety Research Institute (CSRI) is the National Electronic Injury Surveillance System (NEISS). Managed by the U.S. Consumer Product Safety Commission (CPSC), this database serves as a nationally representative probability sample of hospitals in the United States and its territories. NEISS injury data are gathered from the emergency departments (ED) of approximately 100 hospitals selected as a probability sample of all 5,000+ U.S. hospitals with emergency departments. The system's foundation rests on emergency department surveillance data, but the system also has the flexibility to gather additional data at either the surveillance or the investigation level.

Scope of Data

NEISS consumer product related data is available online for the last 20 years. The most recent calendar year of treatment dates generally become available in April of the following year.

Source of Data

URL: https://www.cpsc.gov/cgibin/NEISSQuery/home.aspx

The NEISS Data is collected by the Consumer Product Safety Commission and shared to the public via their website.

NEISS Overview page: https://www.cpsc.gov/Research--Statistics/NEISS-Injury-Data

NEISS FAQ page: https://www.cpsc.gov/Research--Statistics/NEISS-Injury-Data/Neiss-Frequently-Asked-Questions

How is data stored and downloaded?

We can download data from the NEISS portal either as an excel or as a tab limited text or a SAS format. The data can be downloaded from the portal by specifying the data year.

The files downloaded from NEISS, will have the name in the format neissYYYY.xlsx. Each excel file will have data for the year that we downloaded. Below is the list of files that we downloaded from NEISS portal for the year 2005-2024.

['neiss2005.xlsx', 'neiss2006.xlsx', 'neiss2007.xlsx', 'neiss2008.xlsx', 'neiss2009.xlsx', 'neiss2010.xlsx', 'neiss2011.xlsx', 'neiss2012.xlsx', 'neiss2013.xlsx', 'neiss2014.xlsx', 'neiss2015.xlsx', 'neiss2016.xlsx', 'neiss2017.xlsx', 'neiss2018.xlsx', neiss2019.xlsx', 'neiss2020.xlsx', 'neiss2021.xlsx', 'neiss2022.xlsx', 'neiss2023.xlsx', 'neiss2024.xlsx']

2.1.2 Additional Data Sources

None

2.1.3 Data Collection Methodology

NEISS injury data are gathered from the emergency departments (ED) of approximately 100 hospitals selected as a probability sample of all 5,000+ U.S. hospitals with emergency departments. The system's foundation rests on emergency department surveillance data, but the system also has the flexibility to gather additional data at either the surveillance or the investigation level. Surveillance data enable CPSC analysts to make timely national estimates of the number of injuries associated with (but not necessarily caused by) specific consumer products. This data also provides evidence of the need for further study of products. Subsequent follow-back studies yield important clues to the cause and likely prevention of injuries and deaths.

The NEISS data collection methodology initiates at the hospital level, where emergency department staff record injury details during patient admission. A designated NEISS hospital coordinator reviews these daily records to identify cases meeting specific inclusion criteria. The coordinator abstracts pertinent information, translating it into standardized codes based on the NEISS Coding Manual. This coding process captures critical variables, including patient demographics (age, race, ethnicity), injury diagnosis, incident locale, and highly specific product codes that distinguish between similar items (e.g., different types of lawn mowers). A brief narrative description is also added to provide context not captured by codes.

Data entry is performed using CPSC-designed software that applies real-time validation logic to reject inconsistent entries, such as anatomically impossible injuries. Once validated locally, the data is transmitted via a secure internet connection to the CPSC. Upon receipt, the data undergoes a secondary automated editing process before being integrated into the permanent NEISS database. This streamlined workflow allows CPSC analysts in the Directorate for Epidemiology to review the data on the same day it is collected, facilitating immediate quality control and the rapid detection of emerging consumer safety hazards.

2.2 Data Description

2.2.1 Comprehensive Data Dictionary

NEISS provides us two data dictionaries, one for data collected prior to 2018 and another for data collected beyond 2019. Below you can find both the data dictionaries.

Further information about the dataset is available in the 2025 NEISS Coding Manual available online.

Prior to 2018:

Excel/Tab Variable	SAS Variable	Description/Code book
CPSC_Case_Number	NEK	CPSC case number
Treatment_Date	Trmt_date	Date of Treatment
Age	Age	Age of Patient
		0 - Age Not Recorded
		2 to 120 - Age in Years
		201 to 223 - Age 1 to 23 Months
Sex	Sex	Sex of Patient
		0 - Not Recorded
		1 - Male
		2 - Female
Race	Race	Race of Patient
		0 = Not Stated in ED Record
		1 = White
		2 = Black/African American
		3 = Other
		4 = Asian
		5 = American Indian/Alaska Native
		6 = Native Hawaiian/Pacific Islander
Other_Race	Raceoth	Description of Other Race (Used with Race=3)
Body_Part	Bdpt	Injured Body Part
		0 - Internal
		30 - Shoulder
		31 - Upper Trunk
		32 - Elbow
		33 - Lower Arm
		34 - Wrist
		35 - Knee
		36 - Lower Leg
		37 - Ankle
		38 - Pubic Region
		75 - Head
		76 - Face
		77 - Eyeball
		78 - Upper Trunk(Old)
		79 - Lower Trunk
		80 - Upper Arm
		81 - Upper Leg
		82 - Hand
		83 - Foot
		84 - 25-50% of Body
		85 - All Parts Body
		86 - Other(Old)

		87 - Not Stated/Unk
		88 - Mouth
		89 - Neck
		90 - Lower Arm(Old)
		91 - Lower Leg(Old)
		92 - Finger
		93 - Toe
		94 - Ear
Diagnosis	Diag	Injury Diagnosis
		41 - Ingestion
		42 - Aspiration
		46 - Burns, Electrical
		47 - Burns, Not Specified
		48 - Burns, Scald
		49 - Burns, Chemical
		50 - Amputaion
		51 - Burns, Thermal
		52 - Concussions
		53 - Contusions, Abrasions
		54 - Crushing
		55 - Dislocation
		56 - Foreign Body
		57 - Fracture
		58 - Hematoma
		59 - Laceration
		60 - Dental Injury
		61 - Nerve Damage
		62 - Internal Organ Injury
		63 - Puncture
		64 - Strain, Sprain
		65 - Anoxia
		66 - Hemorrhage
		67 - Electric Shock
		68 - Poisoning
		69 - Submersion
		71 - Other/Not Stated
		72 - Avulsion
		73 - Burns, Radiation
		74 - Dermatitis, Conjunctivitis
Other_Diagnosis	Diagoth	Description of Other Diagnosis (Used with Diag=71)
Disposition	Disp	Disposition
		1 - Treated/Examined and Released
		2 - Treated and Transferred
		4 - Treated and Admitted/Hospitalized
		5 - Held for Observation
		6 - Left Without Being Seen

		8 - Fatality, Incl. DOA, Died in ER
		9 - Unknown, Not Stated
Location	Loc	Incident Location
		0 - Not Recorded
		1 - Home
		2 - Farm/Ranch
		4 - Street or Highway
		5 - Other Public Property
		6- Mobile/Manufactured Home
		7 - Industrial
		8 - School/Daycare
		9 - Place of Recreation or Sports
Fire_Involvement	Fmv	Fire Involvement
		0 - No Fire Involved or Fire Involvement Not Recorded
		1 - Fire Involved and Fire Department Attended
		2 - Fire Involved and Fire Department Did Not Attend
		3 - Fire Involved and Unknown Fire Department Attendance
Product_1	Prod1	Product Code (See NEISS Coding Manual for Details)
Product_2	Prod2	Product Code 2 (See NEISS Coding Manual for Details)
Narrative	Narr	Description of Injury Event
Stratum	Stratum	Design Variable-Stratum
PSU	PSU	Design Variable-Primary Sampling Unit (PSU)
Weight	Wt	Statistical Weight for National Estimates

2019 and Beyond:

Excel/Tab Variable	SAS Variable	Description/Code book
CPSC_Case_Number	NEK	CPSC case number
Treatment_Date	Trmt_date	Date of Treatment
Age	Age	Age of Patient
		0 - Age Not Recorded
		2 to 120 - Age in Years
		201 to 223 - Age 1 to 23 Months
Sex	Sex	Sex of Patient
		0 - Not Recorded
		1 - Male
		2 - Female
		3 - Non-Binary/Other (NA before 2021)
Race	Race	Race of Patient
		0 = Not Stated in ED Record

		1 = White
		2 = Black/African American
		3 = Other
		4 = Asian
		5 = American Indian/Alaska Native
		6 = Native Hawaiian/Pacific Islander
Other_Race	Raceoth	Description of Other Race (Used with Race=3)
Hispanic	Hispanic	Hispanic, Latino/Latina, or of Spanish Origin - available from 2019
		0 - Unknown
		1 - Yes
		2 - No
Body_Part	Bdpt	Injured Body Part
		0 - Internal
		30 - Shoulder
		31 - Upper Trunk
		32 - Elbow
		33 - Lower Arm
		34 - Wrist
		35 - Knee
		36 - Lower Leg
		37 - Ankle
		38 - Pubic Region
		75 - Head
		76 - Face
		77 - Eyeball
		79 - Lower Trunk
		80 - Upper Arm
		81 - Upper Leg
		82 - Hand
		83 - Foot
		84 - 25-50% of Body
		85 - All Parts Body
		87 - Not Stated/Unk
		88 - Mouth
		89 - Neck
		92 - Finger
		93 - Toe
		94 - Ear
Body_Part_2	Bdpt2	Injured Body Part 2- available from 2019
Diagnosis	Diag	Injury Diagnosis
		41 - Ingestion

		42 - Aspiration
		46 - Burns, Electrical
		47 - Burns, Not Specified
		48 - Burns, Scald
		49 - Burns, Chemical
		50 - Amputaion
		51 - Burns, Thermal
		52 - Concussions
		53 - Contusions, Abrasions
		54 - Crushing
		55 - Dislocation
		56 - Foreign Body
		57 - Fracture
		58 - Hematoma
		59 - Laceration
		60 - Dental Injury
		61 - Nerve Damage
		62 - Internal Organ Injury
		63 - Puncture
		64 - Strain, Sprain
		65 - Anoxia
		66 - Hemorrhage
		67 - Electric Shock
		68 - Poisoning
		69 - Submersion
		71 - Other/Not Stated
		72 - Avulsion
		73 - Burns, Radiation
		74 - Dermatitis, Conjunctivitis
Diagnosis_2	Diag2	Injury Diagnosis 2 - available from 2019
Other_Diagnosis	Diagoth	Description of Other Diagnosis (Used with Diag=71)
Other_Diagnosis_2	Diagoth2	Description of Other Diagnosis 2 (Used with Diag2=71) - available from 2019
Disposition	Disp	Disposition
		1 - Treated/Examined and Released
		2 - Treated and Transferred
		4 - Treated and Admitted/Hospitalized
		5 - Held for Observation
		6 - Left Without Being Seen
		8 - Fatality, Incl. DOA, Died in ER
		9 - Unknown, Not Stated
Location	Loc	Incident Location

		0 - Not Recorded
		1 - Home
		2 - Farm/Ranch
		4 - Street or Highway
		5 - Other Public Property
		6- Mobile/Manufactured Home
		7 - Industrial
		8 - School/Daycare
		9 - Place of Recreation or Sports
Fire_Involvement	Fmv	Fire Involvement
		0 - No Fire Involved or Fire Involvement Not Recorded
		1 - Fire Involved and Fire Department Attended
		2 - Fire Involved and Fire Department Did Not Attend
		3 - Fire Involved and Unknown Fire Department Attendance
Alcohol	Alcohol	Consumed Alcohol Prior to or During the Incident - available from 2019
		0 - No/No Information
		1 - Yes
Drug	Drug	Use of Drugs or Medication Contributed to the Incident or the Severity of the Injury - available from 2019
		0 - No/No information
		1 - Yes
Product_1	Prod1	Product Code (See NEISS Coding Manual for Details)
Product_2	Prod2	Product Code 2 (See NEISS Coding Manual for Details)
Product_3	Prod3	Product Code 3 (See NEISS Coding Manual for Details) - available from 2019
Narrative	Narr	Description of Injury Event
Stratum	Stratum	Design Variable-Stratum
PSU	PSU	Design Variable-Primary Sampling Unit (PSU)
Weight	Wt	Statistical Weight for National Estimates

2.2.2 Dataset Structure and Description

The dataset provided by NEISS is denormalized conceptually into a star schema with a centralized fact table with injury information and several dimension tables describing different attributes.

2.2.2.1 Fact Table (Transactional Data)

The primary dataset is a file with injury data that is transactional. Each row in the main table represents a unique injury that is getting treated at a hospital. Each record represents injury at a patient level.

Key Information:

Case Number (Identifier-Primary key)

Treatment Date – Date

Demographics (Age, Sex, Race, Other Race) – Coded Integers

Injury Information (Diagnosis, Other Diagnosis, Body Part, Disposition)

Location – Coded Integer

Product Information (Product_1, Product_2, Product_3) – Coded Integers

Narrative Data – Unstructured Data

Fire Involvement – Coded Integer

Statistical Weight – Numeric Value

2.2.2.2 Dimension Tables (Relationships)

The entire NEISS data is present in a denormalized form in an excel file, but the data relies on the logic connecting the NEISS coding manual. The NEISS dataset is encoded and the integer values for the below fields are linked to corresponding description (meaning) from the dimension tables.

Code to Description Relationship

Race: Race is encoded and represents the patients race in the ED record.

Gender: Reports patients gender as in the ED record.

Hispanic: Hispanic variable indicates the ethnicity.

Product Code (Product 1, Product 2, Product 3): A product code is uniquely mapped to the product description. If more than one

product is involved in the injury, the data is available in the product 1, product 2 or product 3 features.

Body Part (Body Part 1 and Body part 2): The body part code is mapped to the body part description. If more than one body part is involved in the injury, corresponding data is available in the body_part_2 feature.

Diagnosis (Diagnosis 1 and Diagnosis 2): The diagnosis code is uniquely mapped to the diagnosis description. If more than one diagnosis is found, the corresponding information is available in the diagnosis_2 feature.

Disposition: Disposition code reports the patient's final disposition as recorded in the ED record.

Location: Location code denotes the location where the injury occurred.

Fire Involvement: Fire involvement code defines if a fire was involved in the injury.

2.2.3 Temporal Coverage and Granularity

The data spans from January 2005 to December 2024 with daily injury records reported to the ED. The coverage of data is 100% across all calendar days of the month between January 2005 to December 2024. The number of injuries seems to have spiked in 2010. We can see a dip in the number of injuries between 2019-2020 due to the COVID19 pandemic and then stabilizing to the pre-pandemic levels.

2.2.4 Data Volume and Scale

Complete dataset contains 7,315,732 transactions with 23 features across 839 products; raw data files total 896.4 MB uncompressed; processing requires approximately 4GB RAM for in-memory analysis.

2.2.5 Categorical Variable Encoding

The below categorical variables are encoded in the NEISS dataset

2.2.5.1 Sex

2.2.5.2 Race/Other Race

2.2.5.3 Hispanic

2.2.5.4 Body Part/Body Part 2

2.2.5.5 Diagnosis/Diagnosis 2

2.2.5.6 Disposition

2.2.5.7 Location

2.2.5.8 Fire Involvement

2.2.5.9 Product 1/Product 2/Product 3

2.3 Data Exploration Report

2.3.1 Univariate Analysis

2.3.2 Bivariate Relationships

2.3.3 Temporal Patterns and Trends

2.3.4 Unexpected Discoveris

2.3.5 Organizational Level Segmentation

2.3.6 Hypothesis Generation

2.4 Data Quality Assessment

2.4.1 Completeness Analysis

2.4.2 Accuracy Assessment

2.4.3 Consistency Evaluation

2.4.4 Timelines and Currency

2.4.5 Relevance and Coverage

2.4.6 Data Quality and Impact Assessment